WAHA NOWEB Engine

Complete Docker Setup Guide — WhatsApp HTTP API for EasySchool

Engine: NOWEB | Image: devlikeapro/waha | Port: 3000

1. What is WAHA & the NOWEB Engine?

WAHA (WhatsApp HTTP API) is a self-hosted server that exposes WhatsApp as a simple REST API. EasySchool calls it from `WahaSmsSender.cs` to send result cards, fee reminders and absent notifications via `/api/sendText` and `/api/sendFile`.

WAHA can run on different **engines**. **NOWEB** is the lightweight engine built on the *Baileys* library — it talks to WhatsApp over a WebSocket directly, **without a browser/Chromium**.

Feature	NOWEB (recommended)	WEBJS
Needs Chromium browser	No	Yes
RAM usage	Low (~150–250 MB)	High (~600 MB+)
Startup speed	Fast	Slower
Best for	Servers / VPS / low-resource hosts	Browser-feature parity

> **Why NOWEB for EasySchool:** lower memory and faster reconnect make it ideal for an always-on messaging server that sends SMS/WhatsApp from background jobs.

2. Prerequisites

- **Docker** installed (Docker Desktop on Windows/Mac, Docker Engine on Linux).
- A free **port 3000** on the host (or pick another and map it).
- A **WhatsApp account on a phone** to scan the pairing QR code.
- Outbound internet access from the server.

3. Install Docker

Windows / macOS

Download & install **Docker Desktop** from `https://www.docker.com/products/docker-desktop`, then start it and confirm the whale icon is running.

Linux (Ubuntu/Debian)

```
curl -fsSL https://get.docker.com -o get-docker.sh
sudo sh get-docker.sh
sudo usermod -aG docker $USER     # log out / back in afterwards
```

Verify the install:

```
docker --version
```

```
docker run --rm hello-world
```

4. Pull the WAHA Image

The free **WAHA Core** image already includes the NOWEB engine:

```
docker pull devlikeapro/waha
```

5. Run WAHA with the NOWEB Engine

A **Quick start** — run a container with NOWEB as the default engine:

```
docker run -d ^
  --name waha ^
  --restart always ^
  -p 3000:3000 ^
  -e WHATSAPP_DEFAULT_ENGINE=NOWEB ^
  -e WAHA_API_KEY=4c4662668dff4e48997492b11d4e7754 ^
  -e WAHA_NOWEB_STORE_ENABLED=true ^
  -e WAHA_NOWEB_STORE_FULLSYNC=true ^
  -v waha-sessions:/app/.sessions ^
  devlikeapro/waha
```

Windows note: the `^` above is the line-continuation character for Windows *CMD*. In **PowerShell** use a backtick `` ` `` instead, and on **Linux/macOS** use a backslash `\`. Or simply put the whole command on one line.

What each setting does

Setting	Purpose
`-p 3000:3000`	Exposes the API on host port 3000.
`WHATSAPP_DEFAULT_ENGINE=NOWEB`	Makes NOWEB the engine for new sessions.
`WAHA_API_KEY`	Protects the API. Clients must send it as the `X-Api-Key` header — exactly what `WahaSmsSender` does.
`WAHA_NOWEB_STORE_ENABLED`	Stores contacts/chats so number lookups work.
`WAHA_NOWEB_STORE_FULLSYNC`	Full history sync on first login (optional).
`-v waha-sessions:/app/.sessions`	**Persists the logged-in session** so you don't re-scan the QR after a restart.
`--restart always`	Auto-restarts the container on reboot/crash.

6. Recommended: docker-compose.yml

For a reproducible, easy-to-restart setup, save this as `docker-compose.yml`:

```
services:
  waha:
    image: devlikeapro/waha
    container_name: waha
    restart: always
    ports:
      - "3000:3000"
    environment:
      WHATSAPP_DEFAULT_ENGINE: NOWEB
      WAHA_API_KEY: "4c4662668dff4e48997492b11d4e7754"
      WAHA_NOWEB_STORE_ENABLED: "true"
      WAHA_NOWEB_STORE_FULLSYNC: "true"
      # Optional: protect the dashboard & swagger UI
      WAHA_DASHBOARD_USERNAME: admin
      WAHA_DASHBOARD_PASSWORD: "change-me"
      WHATSAPP_SWAGGER_USERNAME: admin
      WHATSAPP_SWAGGER_PASSWORD: "change-me"
    volumes:
      - waha-sessions:/app/.sessions

volumes:
  waha-sessions:
```

Then start it:

```
docker compose up -d
docker compose logs -f waha     # watch the logs
```

7. Start a Session & Scan the QR Code

1. Open the **Dashboard**: `http://localhost:3000/dashboard` (or `http://<server-ip>:3000/dashboard`).
2. Open **Swagger** API docs at `http://localhost:3000/`.
3. Create/start a session named **default** (this matches `Waha:Session` in EasySchool).
4. The status becomes `SCAN_QR_CODE` — a QR image appears. On your phone open **WhatsApp** → **Settings** → **Linked Devices** → **Link a device** and scan it.
5. Status changes to `WORKING` — you're connected and ready to send.

You can also do it via API (send the API key header):

```
# Start the "default" session
curl -X POST http://localhost:3000/api/sessions ^
  -H "X-Api-Key: 4c4662668dff4e48997492b11d4e7754" ^
  -H "Content-Type: application/json" ^
  -d "{ \"name\": \"default\", \"start\": true }"

# Fetch the QR code as an image (open the URL in a browser)
http://localhost:3000/api/default/auth/qr?format=image
```

8. Connect EasySchool to WAHA

EasySchool already reads WAHA settings from `School.API/appsettings.json`. Point `BaseUrl` at your container and keep the same API key, session and country code:

```
"Waha": {
  "BaseUrl": "http://localhost:3000",          // or your server URL
  "Session": "default",
  "CountryCode": "92",
  "ApiKey": "4c4662668dff4e48997492b11d4e7754"
}
```

> **Important:** the `ApiKey` in `appsettings.json` must equal the `WAHA_API_KEY` you passed to Docker — otherwise requests get `401 Unauthorized`.

9. Test Sending a Message

Send a test WhatsApp message (replace the number; format is `<countrycode><number>@c.us`):

```
curl -X POST http://localhost:3000/api/sendText ^
  -H "X-Api-Key: 4c4662668dff4e48997492b11d4e7754" ^
  -H "Content-Type: application/json" ^
  -d "{ \"session\": \"default\", \"chatId\": \"923001234567@c.us\", \"text\": \"Hello from
EasySchool \" }"
```

A `200/201` response and a delivered message confirm the full chain *EasySchool → WAHA (NOWEB) → WhatsApp* is working.

10. Useful Management Commands

```
docker ps                       # is WAHA running?
docker logs -f waha             # live logs
docker restart waha             # restart
docker stop waha                # stop
docker compose down             # stop & remove (compose)
docker compose pull && docker compose up -d   # update to latest image
```

11. Troubleshooting

Symptom	Fix
`401 Unauthorized`	API key mismatch — make `WAHA_API_KEY` = `appsettings Waha:ApiKey`.
QR not appearing / session stuck	Restart the session from the dashboard; check `docker logs waha`.
Has to re-scan QR after restart	The `/app/.sessions` volume isn't mounted — add the `-v waha-sessions:/app/.sessions` volume.
Messages fail / number not on WhatsApp	Confirm the number format `92XXXXXXXXXX@c.us` and that the recipient uses WhatsApp.
Port 3000 already in use	Map a different port, e.g. `-p 3100:3000`, and update `BaseUrl`.
Sending too fast / number flagged	EasySchool already throttles (`Waha:Throttle`) — keep delays sane to avoid bans.

> **Production tips:** run behind HTTPS (reverse proxy), set strong dashboard/swagger passwords, back up the `waha-sessions` volume, and use one dedicated WhatsApp number for the server.
